CO3-ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence
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CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM

(a) Representation using Propositional Logic

Let the following propositions be defined:

- F = Patient has Fever
- C = Patient has Cough
- R = Patient has Rash
- B = Patient has Breathlessness
- FL = Patient has Possible Flu
- M = Patient has Possible Measles
- P = Patient has Possible Pneumonia

The given rules can be represented as:

1. Fever AND Cough → Possible Flu
   F ∧ C → FL
2. Fever AND Rash → Possible Measles
   F ∧ R → M
3. Cough AND Breathlessness → Possible Pneumonia
   C ∧ B → P

Facts about Patient A:

- **F = True**
- **C = True**
- **B = True**

Therefore, the initial knowledge base contains:

F(A), C(A), B(A)

(b) Forward Chaining

Forward chaining starts with the known facts and applies rules whose conditions are satisfied.

Step 1:

Known facts:

Fever(A), Cough(A), Breathlessness(A)

Rule 1:

Fever(A) ∧ Cough(A) → PossibleFlu(A)

Since both Fever and Cough are true:

PossibleFlu(A) is derived.

Step 2:

Rule 3:

Cough(A) ∧ Breathlessness(A) → Possible Pneumonia(A)

Since both Cough and Breathlessness are true:

Possible Pneumonia(A) is derived.

Step 3:

Rule 2:

Fever(A) ∧ Rash(A) → Possible Measles(A)

Rash(A) is not given as a fact.

Therefore:

Possible Measles(A) cannot be derived.

Final Conclusions:

- Patient A has Possible Flu.
- Patient A has Possible Pneumonia.
- Possible Measles cannot be concluded.

(c) Backward Chaining for Pneumonia

Goal:

Possible Pneumonia(A)

To prove this goal, identify the rule that concludes Possible Pneumonia:

Cough(A) ∧ Breathlessness(A) → Possible Pneumonia(A)

Therefore, the goal is reduced into two sub-goals:

1. **Cough(A)**
2. **Breathlessness(A)**

Both are already present as facts.

Therefore:

Cough(A) = True
Breathlessness(A) = True

Hence:

Possible Pneumonia(A) = True

Goal Reduction Tree:

Possible Pneumonia(A)
↓
Cough(A) AND Breathlessness(A)
↓ ↓
Cough(A) = True Breathlessness(A) = True
↓ ↓
Fact

Therefore, the goal **"Patient A has Pneumonia" is logically confirmed** by the given knowledge base.

CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT

The FOL knowledge base contains the following rules:

Rule 1:

∀x [Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)]

Rule 2:

∀x [ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x)]

Rule 3:

∀x∀y [Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x)]

Facts:

- Vehicle(Ambulance)
- EmergencyType(Ambulance)
- Vehicle(CarA)
- Behind(CarA, Ambulance)

(a) Unification

Consider Rule 1:

Vehicle(x) ∧ Emergency Type(x) → Clear Path(x)

Given facts:

Vehicle(Ambulance)
EmergencyType(Ambulance)

To match the variable x with Ambulance:

x = Ambulance

Therefore, the substitution set is:

θ = {x/Ambulance}

After substitution:

Vehicle(Ambulance) ∧ EmergencyType(Ambulance) → ClearPath(Ambulance)

Both premises are true.

Therefore:

ClearPath(Ambulance)

is derived.

Now apply Rule 2:

ClearPath(Ambulance) → GreenSignal(Ambulance) ∧ ¬RedSignal(Ambulance)

Therefore:

GreenSignal(Ambulance)

and

¬RedSignal(Ambulance)

are derived.

(b) Resolution Proof for Proceed(CarA)

Goal:

Proceed(CarA)

Relevant rule:

Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x)

Convert into CNF:

¬Vehicle(x) ∨ ¬Behind(x,y) ∨ ¬GreenSignal(y) ∨ Proceed(x)

Given facts:

Vehicle(CarA)

Behind(CarA, Ambulance)

GreenSignal(Ambulance)

Negate the goal:

¬Proceed(CarA)

Resolution Step 1:

Using:

¬Vehicle(CarA) ∨ ¬Behind(CarA,Ambulance) ∨ ¬GreenSignal(Ambulance) ∨ Proceed(CarA)

and:

Vehicle(CarA)

we obtain:

¬Behind(CarA,Ambulance) ∨ ¬GreenSignal(Ambulance) ∨ Proceed(CarA)

Resolution Step 2:

Using:

Behind(CarA,Ambulance)

we obtain:

¬GreenSignal(Ambulance) ∨ Proceed(CarA)

Resolution Step 3:

Using:

GreenSignal(Ambulance)

we obtain:

Proceed(CarA)

Resolution Step 4:

Using the negated goal:

¬Proceed(CarA)

with:

Proceed(CarA)

we derive:

□

The empty clause proves the goal.

Therefore:

Proceed(CarA) is true.

(c) Handling Two Simultaneous Emergency Vehicles

The current knowledge base can handle multiple emergency vehicles only if each emergency vehicle is independently represented by the rules and facts.

However, the knowledge base does not contain rules for:

- Priority between two emergency vehicles.
- Which emergency vehicle should receive the green signal first.
- Conflict resolution between emergency vehicles.
- Simultaneous route allocation.
- Preventing two emergency vehicles from being directed into conflicting paths.

Additional rules can be introduced, such as:

EmergencyPriority(x,y) ∧ HigherPriority(x,y) → GivePriority(x)

and:

GivePriority(x) → GreenSignal(x)

Additional facts should specify the emergency status and priority of each vehicle.

Therefore, the current knowledge base is sufficient for basic emergency routing but requires additional priority and conflict-resolution rules to safely manage two simultaneous emergency vehicles.

CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

Given:

1. **P1: SoilDry → IrrigationNeeded**
2. **P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod**
3. **P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk**

Facts:

SoilDry = True
CropWheat = True

(a) Conversion into CNF

P1:

Original:

SoilDry → IrrigationNeeded

Eliminate implication:

¬SoilDry ∨ IrrigationNeeded

Therefore, CNF is:

(¬SoilDry ∨ IrrigationNeeded)

P2:

Original:

IrrigationNeeded ∧ CropWheat → ApplyDripMethod

Eliminate implication:

¬(IrrigationNeeded ∧ CropWheat) ∨ ApplyDripMethod

Using De Morgan's law:

¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod

Therefore:

(¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod)

P3:

Original:

¬ApplyDripMethod ∧ CropWheat → CropAtRisk

Eliminate implication:

¬(¬ApplyDripMethod ∧ CropWheat) ∨ CropAtRisk

Using De Morgan's law:

ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk

Therefore:

(ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk)

Facts:

SoilDry

and

CropWheat

are already unit clauses in CNF.

Complete CNF:

1. **¬SoilDry ∨ IrrigationNeeded**
2. **¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod**
3. **ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk**
4. **SoilDry**
5. **CropWheat**

(b) Resolution Refutation for ApplyDripMethod

Goal:

ApplyDripMethod

Negate the goal:

¬ApplyDripMethod

Now combine it with the CNF knowledge base.

Resolution Step 1:

From:

¬SoilDry ∨ IrrigationNeeded

and:

SoilDry

derive:

IrrigationNeeded

Resolution Step 2:

From:

¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod

and:

IrrigationNeeded

derive:

¬CropWheat ∨ ApplyDripMethod

Resolution Step 3:

Using:

CropWheat

derive:

ApplyDripMethod

Resolution Step 4:

Using:

ApplyDripMethod

and the negated goal:

¬ApplyDripMethod

derive:

□

Therefore, the theorem is proved:

ApplyDripMethod is true.

Resolution Proof:

SoilDry
↓
IrrigationNeeded
↓
ApplyDripMethod
↓
¬ApplyDripMethod
↓
□

Hence, the knowledge base logically entails **ApplyDripMethod**.

(c) Removing SoilDry

If the fact:

SoilDry

is removed, Rule P1 cannot be activated.

Therefore:

IrrigationNeeded cannot be derived.

Without IrrigationNeeded, Rule P2 cannot derive:

ApplyDripMethod

Thus, there is no basis for deriving either ApplyDripMethod or its negation from the remaining facts.

Rule P3 requires:

¬ApplyDripMethod ∧ CropWheat

Although CropWheat is true, **¬ApplyDripMethod is not given or derived**.

Therefore:

CropAtRisk cannot be logically concluded.

Final conclusion:

If SoilDry is removed:

SoilDry → IrrigationNeeded cannot fire.

Therefore:

IrrigationNeeded → ApplyDripMethod cannot fire.

Since **¬ApplyDripMethod** is also unavailable:

CropAtRisk does not follow.

CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

The agent perceives:

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

Rules:

1. **Stench[1,2] → Wumpus is adjacent to [1,2]**
2. **Breeze[1,1] → Pit is adjacent to [1,1]**
3. **Glitter[x,y] → Gold is at [x,y]**
4. **¬Wumpus[x,y] ∧ ¬Pit[x,y] → Safe[x,y]**

(a) Belief State and Modus Ponens

Initial percepts:

Stench[1,2]

Breeze[1,1]

Glitter[2,2]

Step 1 – Stench

Rule:

Stench[1,2] → Wumpus adjacent to [1,2]

Since Stench[1,2] is true:

Wumpus is adjacent to [1,2]

is derived.

Step 2 – Breeze

Rule:

Breeze[1,1] → Pit adjacent to [1,1]

Since Breeze[1,1] is true:

Pit is adjacent to [1,1]

is derived.

Step 3 – Glitter

Rule:

Glitter[x,y] → Gold[x,y]

For Glitter[2,2]:

Gold[2,2]

is derived.

Derived Knowledge:

1. **Stench[1,2]**
2. **Breeze[1,1]**
3. **Glitter[2,2]**
4. **Wumpus is adjacent to [1,2]**
5. **Pit is adjacent to [1,1]**
6. **Gold is at [2,2]**

The Safe rule cannot be applied because explicit facts **¬Wumpus[x,y]** and **¬Pit[x,y]** are not available.

(b) Forward Chaining

Step 1:

Percept:

Stench[1,2]

Apply Rule 1:

Wumpus adjacent to [1,2]

Step 2:

Percept:

Breeze[1,1]

Apply Rule 2:

Pit adjacent to [1,1]

Step 3:

Percept:

Glitter[2,2]

Apply Rule 3:

Gold[2,2]

Step 4:

Apply Rule 4 only when both conditions are known:

¬Wumpus[x,y] ∧ ¬Pit[x,y]

No such negative facts are provided.

Therefore, no specific cell can be proven safe using Rule 4.

Possible Locations:

From the given simplified rules:

- The Wumpus is known to be **adjacent to [1,2]**.
- A Pit is known to be **adjacent to [1,1]**.
- Gold is confirmed at **[2,2]**.

The exact adjacent cells cannot be uniquely determined from the provided rules because the knowledge base does not specify the complete Wumpus World adjacency structure.

(c) Safety of Path [1,1] → [1,2] → [2,2]

The proposed path is:

[1,1] → [1,2] → [2,2]

At [1,1], the agent has perceived a **Breeze**. Therefore, a Pit is known to be adjacent to [1,1].

At [1,2], the agent has perceived a **Stench**. Therefore, a Wumpus is known to be adjacent to [1,2].

The knowledge base does not provide sufficient negative information such as:

¬Wumpus[1,2]

and

¬Pit[1,2]

Therefore, the Safe rule cannot establish that [1,2] is safe.

Hence, the path:

[1,1] → [1,2] → [2,2]

cannot be logically proven safe using the given knowledge base.

Although the gold at **[2,2]** is confirmed, the agent should not assume that the complete path is safe because the required safety facts are not available.

Final Conclusion:

The path is not guaranteed to be safe.

The agent needs additional percepts or knowledge about the exact locations of the Wumpus and Pit before proceeding.

OVERALL CONCLUSION

The four case studies demonstrate important Artificial Intelligence knowledge representation and reasoning techniques.

- **Case Study 1:** Forward and Backward Chaining can be used for medical diagnosis.
- **Case Study 2:** Unification and Resolution can be used for autonomous traffic management.
- **Case Study 3:** CNF conversion and Resolution Refutation can prove agricultural decisions.
- **Case Study 4:** Modus Ponens and Forward Chaining can help an AI agent reason about the Wumpus World.

These logical reasoning techniques allow intelligent agents to derive valid conclusions from known facts and rules and support effective decision making.